



Zaw Hlaing Htet

Robotic Engineer

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PROFILE

Robotics and Artificial Intelligence Engineer with hands-on experience since 2019 in robotics competitions, embedded systems, and robotic prototyping. Skilled in designing and developing robotic systems, including mechanical design, control systems, kinematics and DH for robotic arm, trouble shooting and hardware–software integration, with practical experience in system troubleshooting and closed-loop control. Experienced in applying AI techniques such as CNN models (YOLO, CSNet) and working with platforms including STM32, Arduino, Raspberry Pi, and MATLAB with Python, C and C++ to build real-world solutions. Demonstrated ability in robotics research, mentoring, and developing practical engineering applications, with a strong passion for innovation and continuous learning in robotics and automation.

WORK EXPERIENCE

Lab Assistant Engineer (Full Time)

Nov 2022 – Dec 2023

Hyper Lab – Yangon

- ◆ Designed and assembled robotics project core kits and external kits for colleges
- ◆ Taught mathematics and 3D modeling such as Blender and Fusion360 to high school students
- ◆ Mentored high school and college students on robotics projects
- ◆ Modelling 3D CAD Mechanical design and wiring assembly for Star City Spray Robot
- ◆ Reversed CAD designed of Robots and Car Parts
- ◆ Assembled, wiring and soldering the components of Drones and Testing
- ◆ Assisted in design of schematics and double layer PCB layout for WIFI UPS System
- ◆ Designed the casing and wiring assembly for mini-UPS for routers
- ◆ Power, Voltage and Current calculation for maximum and minimum Limitation
- ◆ Testing validation for BMS system
- ◆ Assisted in developed and documented plans, assembly, testing and DFM reports

Prototype Engineer (Robotics Internship)

July 2020 – Jan 2021

K Lab – University of Information Technology, Yangon

- ◆ Designed 2D, 3D mechanical design of Delivery Robot for COVID-19 situations for room services
- ◆ Simulation in Virtual Environment such as CoppeliaSim
- ◆ Implemented Emergency message sending and tracking prototype device
- ◆ Developed Water surface cleaning robot prototype
- ◆ Developed and debugged in Keystudio Uno
- ◆ Designed schematics and double layer PCB layout for smart home and humanoid robot prototype
- ◆ Managed Prototype validation testing and contributed process

Research Assistant (Member/Part Time)

Sep 2019 – Feb 2020

Computer Science Department (Yangon University)

- ◆ 3D CAD design and smooth joint movement for Prosthetic Robotic Fingers and wrist
- ◆ Designed the Flexible System with Magnetic wiring Connection at wrist joint
- ◆ Testing MYO muscle sensor with TDD method
- ◆ Measuring dimensions and Collection data for molding and suitable material in Prosthetics
- ◆ Assisted in protype assembly, testing and DFM documentation

EDUCATION

BEng (Hons) Robotics and Artificial Intelligence (1st Class Hons)

Apr 2024 - Apr 2026

University of Hertfordshire

BEng (Hons) Computer Science (Third Year)

Dec 2017 – Mar 2020

University of Yangon

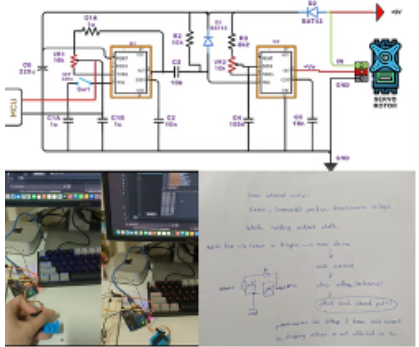
- ◆ Leader of Team TronMM (Robotic Team)
- ◆ Member of Research Team

Research

Experimental Study on Servo Motor Modification and Control Performance Enhancement

I modified a standard servo to improve control flexibility and analyzed its response characteristics such as overshoot and settling time. This helped me better understand real-world control system behavior.

- Investigated limitations of standard servo systems and developed modification approaches to improve flexibility and control performance
- Conducted experimental testing to analyze dynamic response characteristics and system stability
- Applied control system concepts to evaluate and improve system behavior
- Documented findings and performance improvements based on experimental results



Final Year Project (Autonomous MRT Passenger Density Counting and Monitoring System)

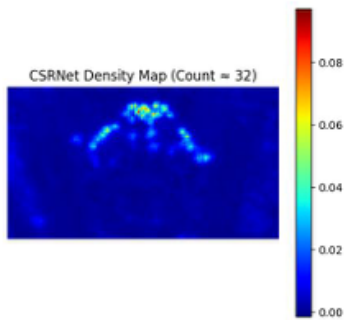
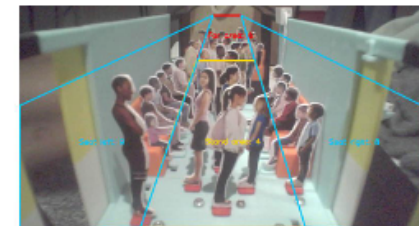
My final year project combines AI-based crowd counting with embedded system integration and control system research, computer vision, and cloud computing to monitor passenger density in train environments.

- Built a **Node.js-based application** for real-time data processing, hardware communication, and system automation
- Developed an **event-driven image capture system** triggered by train door closure, improving data accuracy and reducing unnecessary processing
- Implemented and evaluated multiple AI models (**CSRNet**, **YOLO**, **EfficientNet**) for crowd analysis, selecting CSRNet for high-density performance
- Designed a **privacy-focused cloud pipeline**, transmitting only numerical results to reduce bandwidth and protect user data

Prototype 3D-printed **train carriage model** is designed with **realistic scale**. Door mechanism, Camera positions, Sensor integration and Passenger layout are included. It simulates **real MRT train door operation** with Rack and Pinon gear type.

In this project, the servo motor is modified to provide feedback signals. The modification allows the system to:

- Detect the actual position of the door
- Identify whether the door is **opening, closing, or fully closed**
- **Detect the issues during closing the door (eg, someone is stuck in the door)**
- Generate a digital signal during door movement

A screenshot of the HeadCounter Dashboard. The dashboard shows real-time data and reports. The top section shows the total number of passengers (3) and the number of passengers in the carriage (1). The bottom section shows a table of recent door reports.

Door	Head Count	Counting At	Reported At
2	12	2023-10-10 10:00:00	2023-10-10 10:00:00
2	12	2023-10-10 10:00:00	2023-10-10 10:00:00
2	12	2023-10-10 10:00:00	2023-10-10 10:00:00
2	18	2023-10-10 10:00:00	2023-10-10 10:00:00

Portfolio Projects

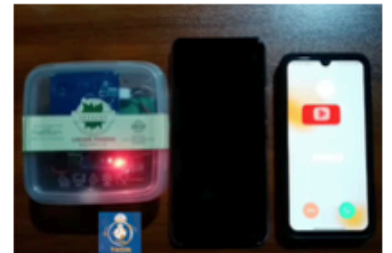
Delivery Robot Prototype for Covid 19 Case

This project is made in during internship. Using Keystudio Uno MCU, this delivery robot prototype is implemented with 9 MG90s servos, 2 DC Gear motor(12V-500RPM), 12V lipo battery. For power supply for MG90s, LM2596 Buck module and for motor, L298n driver. The front 2 wheels are free wheels. For mechanical design, used fusion 360. For IDE software, used Arduino IDE. For wandering, used encoder motor rotation.



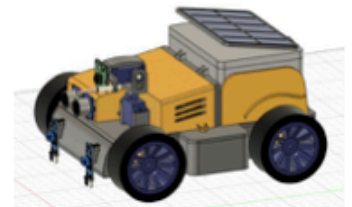
Emergency message sending and tracking device

During the internship, I developed the Emergency message sending and tracking device with SIM 900 module, GPS module and ESP32. This device can send the message including the location for emergency and call the phone that is added in MCU. when user request the device location, user requests the location message using SMS. Then, device calculates the location with GPS and send the location message with SIM900.



Spray Robot 3D CAD design for Star City Yangon

For Smart Garden Project of Star City Yangon, I built the CAD design for prototype and choosing the suitable components. When insects or infection in plants using camera detection, robot sprays the plants. For the tank, bottle Turn-Opener Style is drawn.



Robotic Arm for Industry

For this robotic arm, I drew the System design, mechanical design (CAD) using Fusion 360 included reversed drawing for servos, MCU and Circuits. Kinematics (FK, IK, and DH) and control system are calculated. then, built the algorithm (timestamp and smooth moving) and code in Arduino IDE, and tested the processes.



Mentored Projects

These robotic projects were mentored by me for Science Fair during when I was in job at Hyper Lab and Computer Science Department's Representative Mentor. Tron Luna team got Special Award and Team New York got Second Price and People Choice award.



Crowd Counting with AI Model

Developed a real-time MRT crowd monitoring system using deep learning, implementing EfficientNet-B0 for classification, YOLO for human detection and counting, and CSRNet for dense crowd counting with zone-based analysis. Applied data preprocessing, transfer learning, and model fine-tuning to handle varying crowd densities and improve public transport safety.



Clash of Robot 1 (2018) – People's Choice Award

This competition is held by Myanmar Computer Professionals Association (MCPA) of Government. Participated teams are representation of each university. In that Competition, there are three categories that are obstacles racing, sumo fight, pickup the objects. In these 3 categories, our team (Team Tron) just reached the final and semifinal. However, our team got the People Choice Award.



Science Fair (2018) – Special Award

Project: Drawing Machine Robot

This project innovation show is held by Science Fair organization of University of Yangon. I participated as team leader in Team WMBFI and got the Special award with Drawing Machine (CNC).



First Runner up in Green Planet Challenges

American Center (Yangon) (2019)

This competition is held by Makerfest organization of American Center (Yangon). In that Competition, two categories are included as pick and place for green plants, and hunter with shooting mechanism. Our team (Team Tron) got first runner up.



Clash of Robot 2 (2019) – Winner

This competition is held by Myanmar Computer Professionals Association (MCPA) of Government. Participated teams are representations of 16 universities. In that Competition, there are two categories that are auto sensing the obstacles and part ways, and manual mode for maze solving and racing with balance robots. I participated as Leader and responded in 3D modeling, PID control and Embedded design. In manual control mode, our team (Team Tron) beaten the other team, and got the Champion winner prize.



World Robot Game (Myanmar) 2020

Performance Award

This competition is held by Engineering Association and WRG origination. Our team (Team Tron) participated at Rugby Robot Game among go many other categories. For this Rugby game, four robots are included in one team. I am team leader and response in 3D modeling (CAD), embedded design and control. Our team passed the qualifier and quarter final. at the semifinal, our team lost with 1 marks difference. However, our team got the performance award for Yangon University.



For more details of Projects, Competitions, awards and research are provided in the following website (or Scan QR).

<https://zawhlainghtetcv.wixsite.com/resume-and-portfolio>